

AWS DevOps Engineer Intern Assignment — Documentation Report

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GitHub Repo: [PASTE REPO LINK]
EC2 Public IP: [PASTE PUBLIC IP]

1. Objective

Deploy a static website on an AWS EC2 instance running Ubuntu, served by Nginx, and document the end-to-end process.

2. AWS Services Used

Service	Purpose
EC2	Virtual server (Ubuntu 22.04 LTS, t2.micro free tier)
Security Groups	Virtual firewall — opened ports 22 (SSH) and 80 (HTTP)
Key Pair	RSA key for secure SSH authentication
Elastic IP (bonus)	Static public IP so the address survives instance restarts

3. Linux Commands Used

	Purpose	
date && sudo apt upgrade -y	Update & upgrade packages	
install nginx -y	Install Nginx web server	
ctl status nginx	Check Nginx service status	
ctl restart nginx	Restart Nginx	
	Check disk usage	
	Check memory usage	
-%mem	head`	List top running processes
oem index.html ...	Copy website file to instance	
/var/www/html/	Replace default Nginx page	

4. Steps Performed

1. Launched an Ubuntu 22.04 t2.micro EC2 instance.
2. Created a Security Group allowing SSH (22) and HTTP (80).
3. Connected to the instance over SSH using the downloaded key pair.
4. Updated packages and installed Nginx; verified it was running.
5. Checked disk, memory, and process usage (Linux basics).
6. Uploaded a custom `index.html` and replaced the default Nginx page.
7. Verified the site loads at the EC2 public IP in a browser.
8. Pushed all files + README to GitHub.
9. (Bonus) Attached an Elastic IP / wrote a Nginx restart shell script.

5. Problems Faced & Solutions

Problem	Solution
SSH "Permission denied (publickey)"	Ran <code>chmod 400 key.pem</code> ; used <code>ubuntu@</code> as the user
Website not loading in browser	Port 80 was missing from the Security Group — added it
Default Nginx page still showing after copy	Restarted Nginx after replacing <code>/var/www/html/index.html</code>
<code>.pem</code> too-open-permissions warning on SSH	Fixed key file permissions to <code>400</code>

6. Learnings

- How a Security Group acts as a stateful virtual firewall at the instance level.
- The difference between a dynamic public IP and a static Elastic IP.
- Nginx serves from `/var/www/html/` by default; how `systemctl` manages services.
- Basic Linux server administration: package management, service control, resource monitoring.
- Using SSH key pairs and `scp` for secure remote access and file transfer.

7. Total Time Taken

Approx. [X hours] — EC2 setup (~20 min), Nginx + deployment (~20 min), Git & docs (~30 min).

Deliverables: GitHub repo link, this report (as PDF), EC2 public IP, and screenshots — bundled in a ZIP.